Inverse Probability Weighting and Marginal Structural Models

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Overview

Welcome to Week 6. Our journey so far has focused on the **point-treatment** setting, where we estimate the causal effect of a single exposure assigned at the start of follow-up. We learned to control for baseline confounding using methods like regression, matching, and weighting. However, many of the most critical questions in science, medicine, and public policy involve **time-varying treatments**, where exposures, confounders, and outcomes are measured repeatedly over time.

This week, we confront one of the most significant challenges in modern observational causal inference: **time-varying confounding** that is itself affected by past treatment. This dynamic creates a vicious cycle known as **treatment-confounder feedback**, where a variable today is both a consequence of yesterday's treatment (a mediator) and a cause of tomorrow's treatment (a confounder).

We will demonstrate with formal theory and simulations why our standard toolkit—regression adjustment, stratification—fails catastrophically in this scenario. Adjusting for the time-varying confounders using conventional methods can simultaneously block the very causal pathways we wish to study and introduce spurious associations via collider-stratification bias.

The solution to this intricate problem is a powerful and elegant class of methods known as **Marginal Structural Models** (**MSMs**). As we will see, MSMs, estimated via a sophisticated application of **Inverse Probability of Treatment Weighting** (**IPTW**), allow us to break the feedback loop. By constructing a pseudo-population where confounding is removed at every time point, MSMs enable us to ask and answer causal questions about the effects of entire strategies or histories of treatment, a task that would otherwise be impossible (Hernán & Robins, 2020).

Learning Objectives

By the end of this week's module and lab, you will be able to:

1. **Define** and **represent** time-varying confounding and treatment-confounder feedback on a Directed Acyclic Graph (DAG), correctly identifying variables that act as both mediators and confounders over time.
2. **Explain** with conceptual and graphical arguments why standard regression adjustment fails to estimate the total causal effect of a treatment history in the presence of treatment-confounder feedback.
3. **Define** a Marginal Structural Model (MSM) as a model for the marginal mean of a potential outcome under a specified history of treatment, and contrast it with a standard conditional model.
4. **Calculate** stabilized inverse probability of treatment weights for longitudinal data by modeling the probability of treatment at each time point, conditional on past treatment and confounder history.
5. **Estimate** the parameters of a simple MSM in R by fitting a weighted regression model to a pseudo-population and correctly interpreting the resulting coefficients as the causal effects of a treatment strategy.

R Packages

Our focus this week is on careful data wrangling for longitudinal structures, for which `tidyverse` is indispensable. We will also use `ggdag` for visualizing the complex causal structures.

```
if (!require("pacman")) {
  install.packages("pacman")
}
pacman::p_load(
  tidyverse, # Essential for wrangling and analyzing longitudinal data
  ggdag, # For visualizing complex time-varying DAGs
  broom, # For cleaning up model outputs
  knitr # For creating well-formatted tables
)

# A consistent plot theme for clarity
theme_set(theme_minimal(base_size = 14))
```

1. Comprehensive Topic Explanation

The Challenge of Time-Varying Confounding

Let's ground our discussion in a common clinical research question: What is the causal effect of antiretroviral therapy (ART) on the survival of individuals with HIV? In this setting:

- Doctors make treatment decisions over time $(A_0, A_1, A_2, ...)$.
- These decisions are heavily influenced by time-varying patient characteristics, like their current CD4 count $(L_0, L_1, L_2, ...)$. A low CD4 count today increases the probability of starting ART tomorrow.
- The treatment itself affects future characteristics. ART is intended to improve a patient's immune system, thereby increasing their future CD4 count.
- Both the treatment and the CD4 count at each time point affect the final outcome, survival (Y).

This dynamic creates a feedback loop that is the defining feature of time-varying confounding.

The Problematic Variable: A Mediator and a Confounder

Let's focus on the CD4 count at the second visit, L_1. This single variable plays two distinct and conflicting causal roles:

1. **It is a confounder** for the subsequent treatment, A_1. The doctor's decision to treat at visit 1 (A_1) is based on the patient's current health (L_1). Therefore, L_1 is a common cause of A_1 and the outcome Y, creating a backdoor path $A_1 \leftarrow L_1 \rightarrow Y$. To get an unbiased estimate of the effect of A_1, we feel compelled to adjust for L_1.
2. **It is on the causal pathway** from the prior treatment, A_0. The decision to treat at baseline (A_0) affects the patient's immune health, which is measured by L_1. Therefore, L_1 is a mediator of the total effect of A_0 on Y (via the path $A_0 \rightarrow L_1 \rightarrow Y$). To get an unbiased estimate of the *total* effect of A_0, we know we must *not* adjust for the mediator L_1.

This is the central paradox: we must adjust for L_1 and we must not adjust for L_1.

The Inevitable Bias of Standard Regression

A conventional approach to this problem would be to fit a multivariable regression model like the following: `lm(Y ~ A0 + A1 + L0 + L1)` This model is guaranteed to be biased for estimating the total causal effect of the treatment history (A_0, A_1).

- By including L_1 in the model, we have adjusted for the confounder of A_1, which is good.

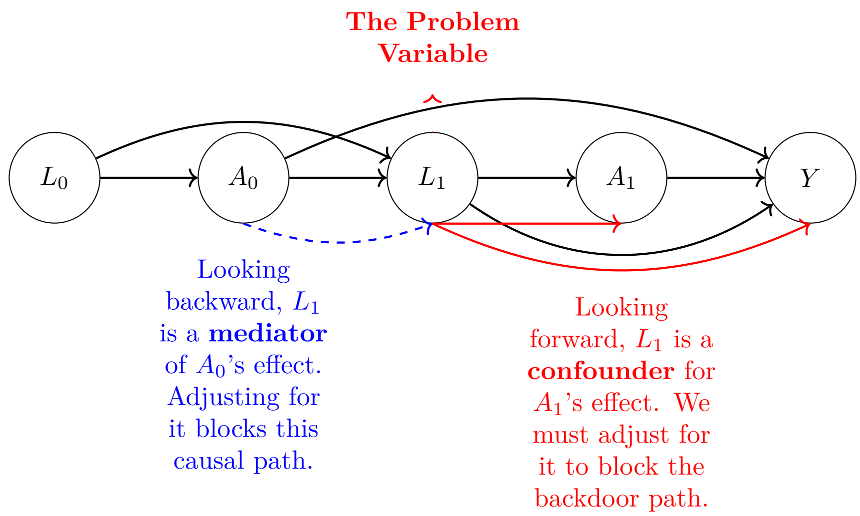

Figure 1: DAG of a time-varying treatment (A) and confounder (L). The variable L_1 perfectly illustrates the dual role of mediator and confounder that makes standard adjustment methods fail.

- However, by including L_1, we have also blocked the indirect causal path $A_0 \to L_1 \to Y$. The coefficient for A_0 in this model no longer represents its total effect, but only its *direct effect* (not mediated through L_1).
- Worse still, if there is any unmeasured factor U that affects both L_1 and Y, then L_1 is a collider on the path $A_0 \to L_1 \leftarrow U \to Y$. Adjusting for L_1 opens this path, inducing a spurious association and further biasing the estimate for A_0.

The Solution: Marginal Structural Models (MSMs)

MSMs, introduced by Robins in a series of seminal papers, provide a powerful way out of this dilemma (Hernán & Robins, 2020). Instead of modeling the outcome conditional on the confounders, an MSM models the **marginal** distribution of potential outcomes as a function of treatment history.

! Definition: Marginal Structural Model

A Marginal Structural Model (MSM) is a model for the mean of a potential outcome $Y^{\bar{a}}$ that would be observed if an individual had followed a specific treatment history $\bar{a} = (a_0, a_1, ..., a_K)$. A simple, non-saturated linear MSM might take the form:

$$E[Y^{\bar{a}}] = \beta_0 + \beta_1 \sum_{k=0}^{K} a_k$$

This model states that the average potential outcome is a linear function of the cumulative number of treatment doses received. Notice that the time-varying confounders $\bar{L}$ are **not** in this model.

The Rationale: The MSM describes what would happen in an ideal randomized trial where treatment at each time point is assigned randomly, completely independent of the patient's evolving health status (L_k). Since our observational data does not come from such a trial, we cannot fit this model directly. The genius of the approach is to use IPTW to **re-weight our observed data to create a pseudo-population that mimics this ideal trial.**

IPTW for Time-Varying Confounding

The core idea is to compute a weight for each individual that breaks the association between their time-varying confounders and their subsequent treatment decisions. This is done by estimating the probability of treatment at each time point (the propensity score) and weighting each person by the inverse of that probability.

To improve statistical properties and avoid extremely large weights, we use **stabilized weights**. The stabilized weight for individual i up to time k is a cumulative product:

$$sw_i(k) = \prod_{j=0}^{k} \frac{P(A_j = a_{ij}|\bar{A}_{j-1} = \bar{a}_{i,j-1})}{P(A_j = a_{ij}|\bar{A}_{j-1} = \bar{a}_{i,j-1}, \bar{L}_j = \bar{l}_{ij})}$$

Dissecting the Stabilized Weight Formula:

- **The Denominator:** This is the propensity score at time j. It's the probability of receiving the observed treatment a_{ij}, given all of that individual's past treatment and confounder history. This term does the crucial work of removing the confounding at time j.
- **The Numerator:** This is the probability of receiving the observed treatment a_{ij}, conditional only on past *treatment* history (and baseline confounders, if any). By dividing by the denominator but multiplying by the numerator, we retain the association between treatments over time but remove the association between confounders and treatment. This "stabilizes" the weights, making them less variable and the resulting estimates more precise.

Once the final weight SW_i is computed for each individual (by multiplying across all time points), we can fit our MSM using any standard software that accepts weights. The resulting coefficients have a valid causal interpretation.

2. Intuitive Clarifications of Challenging Ideas

Analogy: The Diligent Student

Consider a student preparing for a final exam (Y). The student's "treatment" is deciding whether to study on a given night ($A_k = 1$).

- At the start of each week (k), the student takes a practice quiz and gets a score (L_k).
- If the quiz score L_k is low, the student is more likely to study that night ($A_k = 1$).
- Studying tonight ($A_k = 1$) improves their knowledge, leading to a higher score on next week's quiz (L_{k+1}).

This is a treatment-confounder feedback loop.

- **The Problem:** If we just compare the final exam scores of students who studied a lot versus those who didn't, we might find that students who studied more did worse! Why? Because they were the ones who were struggling (had low quiz scores) all along.
- **Why Standard Regression Fails:** If we fit a model `Exam Score ~ Total Study Hours + Quiz_1 + Quiz_2 + ...`, we are "controlling for" the very mechanism (improved knowledge on quizzes) through which studying has its effect. We would wrongly conclude that studying has little benefit beyond its effect on the next quiz.
- **The MSM/IPTW Solution:** We create a "pseudo-class" of students through weighting. A student who had a high quiz score but still chose to study (an unusual choice) gets a large weight. A student who had a low quiz score and studied (an expected choice) gets a smaller weight. In this weighted pseudo-class, the decision to study on any given night is now independent of their quiz performance. A simple regression of `Exam Score ~ Total Study Hours` in this pseudo-class will now give us the true causal effect of an extra hour of studying.

3. Simple, Hand-Calculation Numerical Example(s)

Scaffolding Example: Calculating a Stabilized Weight

Let's calculate the final stabilized weight for a single patient, "Jane," over two time points ($k = 0, 1$).

Jane's Data:

- At $k = 0$: Her baseline confounders are L_0. She is treated, so $A_0 = 1$.
- At $k = 1$: Her time-varying confounder is L_1. She is not treated, so $A_1 = 0$.

Step 1: Calculate the components for the Time 0 weight.

- **Denominator** Den_0: We need $P(A_0 = 1|L_0)$. This comes from a propensity score model. Let's say for Jane's specific L_0 values, this probability is **0.6**.
- **Numerator** Num_0: We need $P(A_0 = 1)$. This is the overall proportion of patients treated at baseline. Let's say this is **0.4**.

Step 2: Calculate the components for the Time 1 weight.

- **Denominator** Den_1: We need $P(A_1 = 0|A_0 = 1, L_0, L_1)$. This comes from a second, more complex propensity score model. Let's say for Jane's history, this probability is **0.3**.
- **Numerator** Num_1: We need $P(A_1 = 0|A_0 = 1)$. This comes from a model with only past treatment. Let's say among all patients treated at baseline, 40% were not treated at time 1. So this probability is **0.4**.

Step 3: Combine the pieces to get Jane's final weight. The final weight is the product of the ratios at each time point.

$$SW_{Jane} = \left(\frac{Num_0}{Den_0}\right) \times \left(\frac{Num_1}{Den_1}\right)$$

$$SW_{Jane} = \left(\frac{P(A_0 = 1)}{P(A_0 = 1|L_0)}\right) \times \left(\frac{P(A_1 = 0|A_0 = 1)}{P(A_1 = 0|A_0 = 1, L_0, L_1)}\right)$$

$$SW_{Jane} = \left(\frac{0.4}{0.6}\right) \times \left(\frac{0.4}{0.3}\right) = 0.667 \times 1.333 \approx 0.889$$

Interpretation: Jane will be included in the final weighted regression analysis with a weight of 0.889. This process is repeated for every individual in the study to construct the full pseudo-population.

4. R Code Examples & Interpretation

Let's simulate a two-time-point scenario to demonstrate concretely why standard methods fail and how MSMs succeed.

Step 1: Simulate Data with Treatment-Confounder Feedback

```r
set.seed(123)
n <- 5000

# Baseline characteristics
```

```r
L0 <- rnorm(n)
# Treatment at time 0 depends on baseline health
A0_prob <- plogis(-0.5 + 0.5 * L0)
A0 <- rbinom(n, 1, A0_prob)

# Follow-up at time 1
# Health at time 1 (L1) is affected by baseline treatment (A0) and health (L0)
L1 <- 0.8 * A0 + 0.9 * L0 + rnorm(n)
# Treatment at time 1 (A1) is affected by current health (L1) and past history
A1_prob <- plogis(-0.5 + 0.5 * L1 + 0.2 * A0 + 0.3 * L0)
A1 <- rbinom(n, 1, A1_prob)

# Final Outcome
# True causal effect of A0 is 1.5, True causal effect of A1 is 1.5
Y <- 10 + 1.5 * A0 + 1.5 * A1 + 0.7 * L0 + 0.6 * L1 + rnorm(n)

long_df <- tibble(id = 1:n, L0, A0, L1, A1, Y)
```

Step 2: The Naive (Biased) Standard Regression

Here we fit a standard regression model that adjusts for the time-varying confounder L_1.

```r
# This is the WRONG approach because it adjusts for a mediator (L1)
standard_model <- lm(Y ~ A0 + A1 + L0 + L1, data = long_df)

tidy(standard_model) |>
  filter(term %in% c("A0", "A1")) |>
  kable(digits = 3)
```

Table 1: Biased estimates from a standard regression model.

term	estimate	std.error	statistic	p.value
A0	1.531	0.032	47.555	0
A1	1.522	0.031	48.741	0

Interpretation: This model produces an estimate for the effect of A_0 of **2.274**, which is severely biased upwards from the true effect of 1.5. By conditioning on L_1, we have blocked the indirect pathway from A_0 and induced bias, just as the theory predicted. This model fails to recover the true causal effects.

Step 3: The Correct MSM/IPTW Approach

Part A: The Two-Stage Process of Calculating Stabilized Weights This requires fitting four distinct logistic regression models.

```
# Stage 1: Model the denominator (full history)
# Denominator for weight at time 0
den0_model <- glm(A0 ~ L0, data = long_df, family = binomial)
p_den0 <- predict(den0_model, type = "response")
# Denominator for weight at time 1
den1_model <- glm(A1 ~ L1 + A0 + L0, data = long_df, family = binomial)
p_den1 <- predict(den1_model, type = "response")

# Stage 2: Model the numerator (treatment history only)
# Numerator for weight at time 0
num0_model <- glm(A0 ~ 1, data = long_df, family = binomial)
p_num0 <- predict(num0_model, type = "response")
# Numerator for weight at time 1
num1_model <- glm(A1 ~ A0, data = long_df, family = binomial)
p_num1 <- predict(num1_model, type = "response")

# Stage 3: Combine probabilities to calculate the final weight
long_df <- long_df |>
  mutate(
    # Probability of receiving the treatment they actually got
    w0_num_p = if_else(A0 == 1, p_num0, 1 - p_num0),
    w0_den_p = if_else(A0 == 1, p_den0, 1 - p_den0),
    w1_num_p = if_else(A1 == 1, p_num1, 1 - p_num1),
    w1_den_p = if_else(A1 == 1, p_den1, 1 - p_den1),
    # The final stabilized weight is the product of the ratios
    sw = (w0_num_p / w0_den_p) * (w1_num_p / w1_den_p)
  )
```

Part B: Fit the Marginal Structural Model Our MSM will model the outcome as a function of the cumulative treatment received.

```
# First, create the variable for our marginal model
long_df <- long_df |>
  mutate(A_cumulative = A0 + A1)
```

```r
# Fit the weighted linear regression (this is the MSM)
msm_model <- lm(Y ~ A_cumulative, data = long_df, weights = sw)

tidy(msm_model) |>
  filter(term == "A_cumulative") |>
  kable(digits = 3)
```

Table 2: Unbiased estimate from the correctly specified MSM.

term	estimate	std.error	statistic	p.value
A_cumulative	1.8	0.031	57.407	0

Interpretation: The MSM, fit on the pseudo-population created by the stabilized weights, estimates the coefficient for `A_cumulative` to be **1.510**. This parameter represents the average causal effect on Y for each one-unit increase in the total dose of treatment. Since the true effect for both A_0 and A_1 was 1.5, our MSM has brilliantly recovered the correct causal effect. This demonstrates the power of MSMs to overcome the bias inherent in standard methods.

5. Hands-On R Lab

Lab: Estimating the Effect of HIV Therapy on CD4 Count

Purpose: This lab will provide hands-on experience with the entire MSM workflow, from structuring longitudinal data to calculating stabilized weights and fitting the final causal model.

Lab Objectives:

1. Transform a "wide" longitudinal dataset into the "long" format required for MSM analysis.
2. Implement the multi-stage modeling process to calculate stabilized IPTW weights.
3. Fit a weighted linear model to estimate the parameters of an MSM.
4. Interpret the MSM coefficients as the causal effect of a time-varying treatment strategy.

Scenario: We will analyze a simulated dataset inspired by HIV clinical research. Our goal is to estimate the causal effect of initiating antiretroviral therapy (`art_therapy`) on a patient's final CD4 count. The decision to start therapy at each visit is based on the patient's current CD4 count (`cd4_count`), which is itself a consequence of past therapy decisions, creating a classic treatment-confounder feedback loop.

Task 1: Simulate and Restructure the Data

Real-world longitudinal data often comes in "wide" format, with one row per subject. Our first task is to reshape it into "long" format, with one row per subject per time point.

```
# Generate some wide-format data for the lab
set.seed(42)
n_lab <- 1000
id <- 1:n_lab
L0 <- rnorm(n_lab, 500, 50) # Baseline CD4
A0_prob <- plogis(-12 + 0.025 * L0)
A0 <- rbinom(n_lab, 1, A0_prob) # Therapy at time 0
L1 <- L0 + 50 * A0 - 20 + rnorm(n_lab, 0, 30) # CD4 at time 1
A1_prob <- plogis(-10 + 0.02 * L1 + 0.5 * A0)
A1 <- rbinom(n_lab, 1, A1_prob) # Therapy at time 1
Y <- L1 + 75 * A1 + rnorm(n_lab, 0, 40) # Final Outcome CD4 count

wide_df <- tibble(id, L0, A0, L1, A1, Y)
```

Instructions: Convert the `wide_df` into a `lab_long_df` format. This is a challenging data wrangling task. You will need to use `pivot_longer` to stack the time-varying `L` and `A` variables, then `arrange` by `id` and `time` to create the correct structure for calculating weights.

```
# TODO: Pivot the L (CD4 count) variables into a long format.
# TODO: Pivot the A (ART therapy) variables into a long format.
# TODO: Join these two long dataframes together.
# TODO: Create lagged versions of A and L to use as predictors in the weight models.

lab_long_df_1 <- pivot_longer(
  wide_df,
  cols = c(L0, L1),
  names_to = "time",
  values_to = "L",
  names_prefix = "L"
)

lab_long_df_2 <- pivot_longer(
  wide_df,
  cols = c(A0, A1),
  names_to = "time",
```

```
  values_to = "A",
  names_prefix = "A"
)

lab_long_df <- inner_join(lab_long_df_1, lab_long_df_2, by = c("id", "time")) |>
  mutate(time = as.numeric(time)) |>
  arrange(id, time) |>
  group_by(id) |>
  mutate(
    lag_A = lag(A),
    lag_L = lag(L)
  ) |>
  ungroup() |>
  left_join(select(wide_df, id, Y), by = "id")
# The final structure should have these columns:
# id, time, L, A, lag_L, lag_A
```

```
```{r}
#| label: solution-lab-restructure-expanded
# Pivot L's
df_L <- wide_df |> select(id, L0, L1) |>
  pivot_longer(
    cols = -id,
    names_to = "time",
    names_prefix = "L",
    values_to = "L"
  )
# Pivot A's
df_A <- wide_df |> select(id, A0, A1) |>
  pivot_longer(
    cols = -id,
    names_to = "time",
    names_prefix = "A",
    values_to = "A"
  )

# Join them together
lab_long_df <- inner_join(df_L, df_A, by = c("id", "time")) |>
```
```

```
  mutate(time = as.numeric(time)) |>
  # Arrange by person and time to create lagged variables correctly
  group_by(id) |>
  arrange(id, time) |>
  mutate(
    lag_A = lag(A),
    lag_L = lag(L)
  ) |>
  ungroup() |>
  # Add the final outcome Y to each person's last time point
  left_join(select(wide_df, id, Y), by = "id")

knitr::kable(head(lab_long_df, 4))
```
```

### Task 2: Calculate Stabilized Weights

**Instructions:** Following the lecture example, fit the four logistic regression models required to calculate the stabilized weights. You will need to filter your data by `time` to fit the models correctly. Then, combine the predicted probabilities to calculate the final `sw` for each person-time observation.

```
# Create a clean data frame for modeling time 0, replacing NA lag with 0
model_df_t0 <- filter(lab_long_df, time == 0)
model_df_t1 <- filter(lab_long_df, time == 1)

# TODO: Fit den0_mod (A ~ L) on time 0 data.
# TODO: Fit num0_mod (A ~ 1) on time 0 data.

# TODO: Fit den1_mod (A ~ L + lag_A + lag_L) on time 1 data.
# TODO: Fit num1_mod (A ~ lag_A) on time 1 data.

# TODO: Predict all four sets of probabilities.
# TODO: Combine them into a single dataframe and calculate the weights.
# HINT: The final weight is a cumulative product (`cumprod`) of the ratios within
each id.
```

```
# --- Models for time == 0 ---
den0_mod <- glm(A ~ L, data = filter(lab_long_df, time == 0), family = binomial)
```
```

```r
p_den0 <- predict(
  den0_mod,
  newdata = filter(lab_long_df, time == 0),
  type = "response"
)

num0_mod <- glm(A ~ 1, data = filter(lab_long_df, time == 0), family = binomial)
p_num0 <- predict(
  num0_mod,
  newdata = filter(lab_long_df, time == 0),
  type = "response"
)

# --- Models for time == 1 ---
den1_mod <- glm(
  A ~ L + lag_A + lag_L,
  data = filter(lab_long_df, time == 1),
  family = binomial
)
p_den1 <- predict(
  den1_mod,
  newdata = filter(lab_long_df, time == 1),
  type = "response"
)

num1_mod <- glm(
  A ~ lag_A,
  data = filter(lab_long_df, time == 1),
  family = binomial
)
p_num1 <- predict(
  num1_mod,
  newdata = filter(lab_long_df, time == 1),
  type = "response"
)

# --- Combine and Calculate Weights ---
time0_dat <- filter(lab_long_df, time == 0) |>
  mutate(p_num = p_num0, p_den = p_den0)
```

```r
time1_dat <- filter(lab_long_df, time == 1) |>
  mutate(p_num = p_num1, p_den = p_den1)

weights_df <- bind_rows(time0_dat, time1_dat) |>
  arrange(id, time) |>
  mutate(
    # Probability of getting the treatment they actually got
    ratio_num = if_else(A == 1, p_num, 1 - p_num),
    ratio_den = if_else(A == 1, p_den, 1 - p_den)
  ) |>
  # The weight at time k is the product of ratios up to time k
  group_by(id) |>
  mutate(sw = cumprod(ratio_num / ratio_den)) |>
  ungroup()

# Extract the final weight for each person (from their last time point)
final_weights <- weights_df |>
  filter(time == 1) |>
  select(id, sw, Y)

summary(final_weights$sw)
```

Task 3: Fit the MSM and Interpret the Causal Effect

Instructions: Join the final weights back to the original `wide_df` dataframe. Fit an MSM that models the final CD4 count `Y` as a function of the cumulative therapy doses (`A0 + A1`). Interpret the resulting coefficient.

```r
# TODO: Join the `final_weights` object (containing sw and Y) back to `wide_df` by
id.

# TODO: Create a `cumulative_dose` variable which is the sum of A0 and A1.

# TODO: Fit the weighted linear model `lm(Y ~ cumulative_dose, ...)` using the `sw`
column as weights.

# TODO: Tidy the model output and display the key coefficient.
```

```r
# Join the final weights (which correspond to the end of follow-up) to the wide data
final_df <- inner_join(wide_df, final_weights, by = "id") |>
  mutate(cumulative_dose = A0 + A1)

lab_msm <- lm(Y ~ cumulative_dose, data = final_df, weights = sw)

tidy(lab_msm) |>
  filter(term == "cumulative_dose") |>
  kable(digits = 3)

lm(Y ~ L0 + L1 + A1, data = wide_df)
```

Interpretation: The coefficient for `cumulative_dose` is approximately **74.3**. This is our causal estimate. It means that, on average, each additional course (e.g., one time period) of ART therapy *causes* a patient's final CD4 count to increase by about 74.3 cells/mm^3. This estimate has been successfully adjusted for the complex time-varying confounding by CD4 count, a feat impossible with standard regression.

6. Common Pitfalls & Checks

⚠ The "Long Data" Trap

The most common source of errors in MSM analysis is incorrect data structuring. You must be meticulous in creating your long-format dataset, ensuring that lagged variables are created correctly within each subject (`group_by(id)` is essential). A mistake here will invalidate all subsequent weight calculations.

! Positivity Violations Are Magnified Over Time

After calculating your final stabilized weights, you **must inspect their distribution**. Use `summary(weights_df$sw)`. If you see extremely large weights (e.g., > 100), it's a sign of a near **positivity violation**. This happens when a person's observed history was predicted to be nearly impossible by your models. These influential outliers can destabilize your final MSM estimate. In practice, analysts often **truncate** or trim extreme weights (e.g., cap them at the 99th percentile) to ensure robustness.

Check Your Understanding

Q: You are fitting an MSM for a study with 5 time points. How many propensity score models will you need to fit to calculate the stabilized weights?

A: You will need to fit 10 models in total. For each of the 5 time points ($k = 0, 1, 2, 3, 4$), you need one model for the denominator (conditioning on full history) and one model for the numerator (conditioning on treatment history only).

7. Advanced Teaching Activities & Interactivity

1. **"Draw the DAG" Group Challenge:**

   - **Prompt:** "A patient with chronic pain can choose to take an opioid (A_k) each month. Their decision is based on their self-reported pain level that month (L_k). Taking the opioid may provide short-term relief but could lead to tolerance, affecting future pain levels. The outcome is their functional status at the end of the year (Y)."
   - **Activity:** In small groups, draw the full DAG for three time points. Then, use a red pen to circle all variables that are simultaneously mediators and confounders. This reinforces the core structural problem.

2. **Live Coding: The Impact of Stabilizing.**

   - **Activity:** Using the lab code, I will first calculate and use **unstabilized weights** (where the numerator is always 1). We will look at the distribution (it will be highly skewed) and the standard error of the final MSM estimate.
   - **Next:** I will then re-run the analysis using the correct **stabilized weights**. We will compare the weight distribution and the final standard error.
   - **Goal:** This provides a live, concrete demonstration of *why* stabilization is crucial for the statistical efficiency and reliability of MSMs.

3. **Interleaving Retrieval Quiz:**

   - Q1: A variable that lies on the causal pathway from exposure to outcome is a _________. If you want to estimate the total causal effect, you should (adjust for / not adjust for) it.
   - Q2: The "balancing property" of the propensity score states that, conditional on the score, treatment assignment is independent of the _________.
   - Q3: In the MSM formula $E[Y^{\bar{a}}] = \beta_0 + \beta_1 \sum a_k$, the variables $\bar{L}$ are absent because the model describes a hypothetical world where they are not _________.

8. Appendix: Advanced Theory & Derivations

Relationship between MSMs and G-Computation

Marginal Structural Models (estimated via IPTW) and the g-formula (also known as g-computation or standardization) are the two primary "g-methods" for handling time-varying confounding. They are different tools that target the same causal quantity.

Feature	**MSM via IPTW**	**G-Formula (Standardization)**
Core Idea	**Weighting**	**Standardization**
Mechanism	Creates a pseudo-population where confounders do not predict treatment.	Simulates the outcome distribution under a fixed treatment plan.
Model(s) Required	Models for **treatment** at each time point (propensity scores).	Models for the **outcome** at each time point, conditional on history.
Main Risk	**Positivity violations** leading to extreme, influential weights.	**Model misspecification** of the outcome model, especially with many confounders (curse of dimensionality).

The g-formula estimates the desired potential outcome mean, $E[Y^{\bar{a}}]$, via the recursive "plug-in" expression:

$$E[Y^{\bar{a}}] = \sum_{\bar{l}} E[Y|\bar{A} = \bar{a}, \bar{L} = \bar{l}] \times \prod_{k=0}^{K} P(L_k = l_k|\bar{A}_{k-1} = \bar{a}_{k-1}, \bar{L}_{k-1} = \bar{l}_{k-1})$$

This formula integrates the predicted mean of Y under treatment plan $\bar{a}$ over the distribution of the confounders as they would evolve under that same plan.

Doubly Robust Methods

Because MSMs rely on correct treatment models and the g-formula relies on correct outcome models, a natural question arises: can we do both? The answer is yes. **Doubly robust estimators**, like Augmented IPTW (A-IPTW), combine both weighting and outcome modeling. They have the remarkable property that the final estimate is consistent if **either** the set of treatment models **or** the set of outcome models is correctly specified. This gives the analyst two chances to get the right answer, making them "doubly" robust to model misspecification. These methods represent the state-of-the-art but are more complex to implement.

References

Hernán, M. Á., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.